

The answer key for the **Mastery** round is below.

p	Answer	p	Answer
1	26	5	230
2	10	6	518
3	$\frac{3}{4}$	7	$\frac{4\pi+2+\sqrt{3}}{4}$
4	-1	8	76

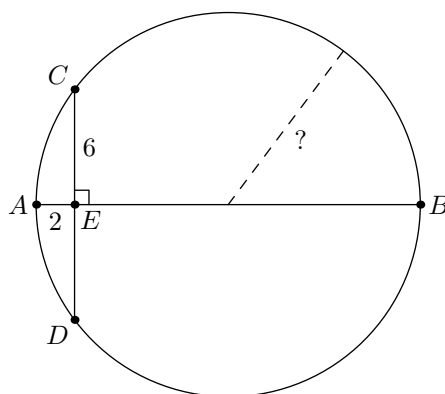
Problem 1. Find the least positive integer x such that $x - 1$ is divisible by 5, and $x - 5$ is divisible by 7.

Proposed by: Zack Berchenko

Answer: 26

Solution: For $x - 5$ to be divisible by 7, x must be one of 5, 12, 19, 26, The first of these that satisfy $x - 1$ being divisible by 5 is 26.

Problem 2. In a circle, AB is a diameter and CD is a chord perpendicular to AB , intersecting AB at point E . Given that $CE = 6$ and $AE = 2$, find the radius of the circle.



Proposed by: Taison Scofield

Answer: 10

Solution: Let r be the radius of the circle.

Since AB is a diameter and $CE \perp AB$, AB bisects CD , so $DE = CE = 6$. Now, observe that $AB = 2r$, so $BE = 2r - 2$. As such, by Power of a Point, we have $AE \cdot BE = 4r - 4 = CE \cdot DE = 36$, so $r = \span style="border: 1px solid black; padding: 0 5px;">10.$

Problem 3. Austin orders two Sloppy Joe sandwiches from Bubway. Each of Bubway's sandwiches has a random size, which is a real number chosen uniformly at random between 0 and 1. After Austin gets his first sandwich, he guesses if it will be bigger or smaller than his second one. If he guesses optimally, compute the probability he is right.

Proposed by: Michael Luo

Answer: 3
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4

Solution: Clearly Austin should guess that the sandwich is bigger if the size is at least $\frac{1}{2}$, and guess the sandwich is smaller if the size is less than $\frac{1}{2}$.¹

Assume the sandwich has a size of less than $\frac{1}{2}$, the case where the size is bigger is analogous. If the size is $s < \frac{1}{2}$, then Austin guesses that the sandwich is smaller. The probability he is wrong is s . So averaging over all $0 < s < \frac{1}{2}$, by geometric probability, the probability he is wrong is $\frac{1}{4}$. So the probability he is right is $\frac{3}{4}$.

Remark. This problem is a *much* easier version of the following problem, which was deemed too difficult for contest.

¹There is the edge case where the size is exactly $\frac{1}{2}$, but this occurs with probability 0.

Austin is ordering Sloppy Joe sandwiches from Subway. Each of Subway's sandwiches has a random size, which is a real number chosen uniformly at random between 0 and 1.

Austin has three lunchboxes, labeled 1, 2, and 3. He does the following three times: he orders a sandwich, then he places it in an empty lunchbox. At the end of this procedure, let s_i be the size of the sandwich in lunchbox i . Austin places the sandwiches in the lunchboxes to maximize the probability $s_1 < s_2 < s_3$.

Compute the probability $s_1 < s_2 < s_3$.

Here's a solution sketch for the harder problem. Let $f(n)$ be the probability we can correctly order n sandwiches, so the original problem is to compute $f(2)$, and the harder problem is to compute $f(3)$. The key observation is the recurrence relation

$$f(n+1) = \int_0^1 \max_{0 \leq i \leq n} \left(\binom{n}{i} (1-s)^i s^{n-i} f(i) f(n-1-i) \right) ds.$$

We get $f(0) = f(1) = 1$, so $f(2) = \frac{3}{4}$ and $f(3) = \frac{377}{726}$.

Additionally,

$$f(4) = \frac{1037891564126671}{3020778029791104} \approx 0.343584187216,$$

and

$$f(5) \approx 0.221406029898.$$

When written as a rational number in simplest form, the numerator of $f(5)$ is on the order of $3.0043741074 \cdot 10^{95}$. It did not fit on the page.

Problem 4. Let $r < s < t$ be the real roots of $x^3 + 14x^2 - 34$. Compute $\frac{s}{r} + \frac{s}{t}$.

Proposed by: Jefferson Zhou

Answer: -1

Solution: Observe that $\frac{s}{r} + \frac{s}{t} = \frac{rs + st}{rt}$. Also, by Vieta's, $rs + st + rt = 0$, so now, $\frac{rs + st}{rt} = \frac{-rt}{rt} = \span style="border: 1px solid black; padding: 2px;">-1.$

Problem 5. Let the remainder when 7^{777} is divided by 1000 be p , and the remainder when 3^{333} is divided by 100 be q . Find $p + q$.

Proposed by: Taison Scofield

Answer: 230

Solution: First, by Euler's theorem, since $\phi(100) = 40$, we have $3^{333} \equiv 3^{13} \pmod{100}$. We compute $3^1 = 3$, $3^2 = 9$, $3^4 = 81$, $3^8 = 61$, so $3^{13} = 3^8 \cdot 3^4 \cdot 3 = 61 \cdot 81 \cdot 3$. Now $61 \cdot 81 = 4941$, and $4941 \cdot 3 = 14823 \equiv 23 \pmod{100}$, so $q = 23$.

Next, we compute $p = 7^{777} \pmod{1000}$ using the Chinese Remainder Theorem. First, mod 8: since $7 \equiv -1 \pmod{8}$, we get $7^{777} \equiv (-1)^{777} \equiv -1 \equiv 7 \pmod{8}$. Then, mod 125: since $\phi(125) = 100$, Euler gives $7^{777} \equiv 7^{77} \pmod{125}$.

We compute powers of 7 mod 125: $7^1 = 7$, $7^2 = 49$, $7^4 = 26$, $7^8 = 51$, $7^{16} = 101$, $7^{32} = 76$, $7^{64} = 26$.

Now write $77 = 64 + 8 + 4 + 1$, so $7^{77} = 7^{64} \cdot 7^8 \cdot 7^4 \cdot 7 = 26 \cdot 51 \cdot 26 \cdot 7$. $26 \cdot 51 = 1326 \equiv 76$, then $76 \cdot 26 = 1976 \equiv 101$, and $101 \cdot 7 = 707 \equiv 82 \pmod{125}$.

So $p \equiv 82 \pmod{125}$ and $p \equiv 7 \pmod{8}$. Let $p = 125k + 82$. Then $125k + 82 \equiv 7 \pmod{8} \Rightarrow 5k + 2 \equiv 7 \Rightarrow 5k \equiv 5 \Rightarrow k \equiv 1 \pmod{8}$, so $k = 8m + 1$. Then $p = 125(8m + 1) + 82 = 1000m + 207 \Rightarrow p \equiv 207 \pmod{1000}$.

Now $p = 207$ and $q = 23$, so the final answer is $p + q = \boxed{230}$. Note: It is also possible to use the Binomial Theorem.

Problem 6. Markov is climbing up a chain. The chain is 35 feet long, and Markov starts at the very bottom of the chain. Every 8 feet, Markov has a $\frac{n}{40}$ chance of falling off the chain and restarting from the bottom, where n is the number of feet up the chain Markov is. If Markov can climb at a rate of 1 foot per minute, what is the expected amount of minutes it takes Markov to climb to the top of the chain?

Proposed by: Zack Berchenko

Answer: $\boxed{518}$

Solution: The expected number of feet Markov climbs per run is

$$8 + 8 \cdot \frac{4}{5} + 8 \cdot \frac{4}{5} \cdot \frac{3}{5} + 8 \cdot \frac{4}{5} \cdot \frac{3}{5} \cdot \frac{2}{5} + 3 \cdot \frac{4}{5} \cdot \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{5}.$$

The probability of any run succeeding is

$$\frac{4}{5} \cdot \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{5},$$

so the expected number of runs is

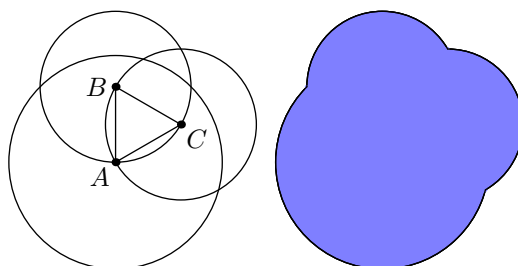
$$\frac{1}{\frac{4}{5} \cdot \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{5}}.$$

Now the expected number of feet in total is the expected number of runs times the expected number of feet per run, which is

$$\frac{8 + 8 \cdot \frac{4}{5} + 8 \cdot \frac{4}{5} \cdot \frac{3}{5} + 8 \cdot \frac{4}{5} \cdot \frac{3}{5} \cdot \frac{2}{5} + 3 \cdot \frac{4}{5} \cdot \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{5}}{\frac{4}{5} \cdot \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{5}} = \frac{625}{3} + \frac{500}{3} + 100 + 40 + 3 = 518,$$

so the expected number of minutes is $\boxed{518}$.

Problem 7. In equilateral triangle ABC with side length $\frac{\sqrt{2}}{2}$, a circle of radius 1 is drawn with center A and circles of radius $\frac{\sqrt{2}}{2}$ are drawn with centers B and C . Compute the area of the union of the 3 circles, which is shaded in the diagram below.



Proposed by: Jefferson Zhou

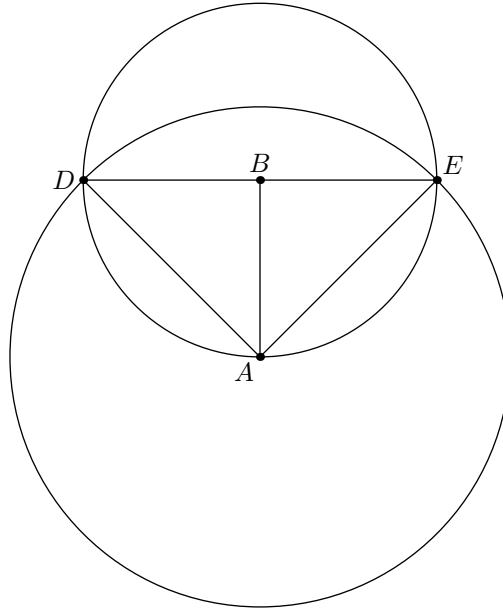
Answer: $\boxed{\frac{4\pi + 2 + \sqrt{3}}{4}}$

Solution: (Note: I'm using the same name for the region contained by a circle and the perimeter of the circle interchangeably, hopefully it is clear by context what is meant in each case.)

We will use PIE on this problem. First, note that the sum of the areas of the three circles is $\left(1 + \frac{1}{2} + \frac{1}{2}\right)\pi = 2\pi$.

We now need to find the areas of the pairwise intersections of the circles. Let ω_A , ω_B , and ω_C be the circles centered at A , B , and C , respectively.

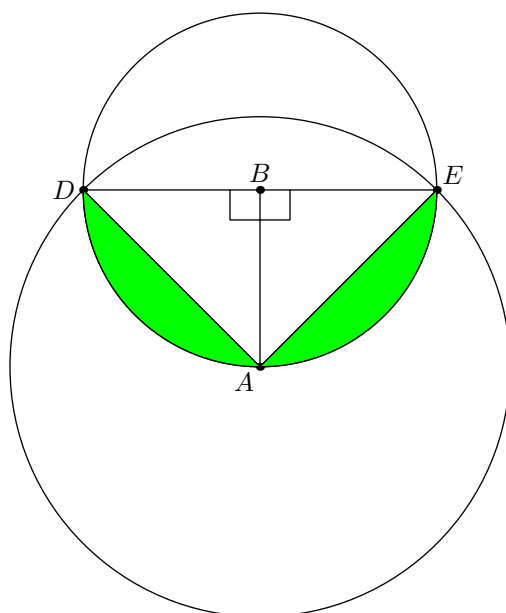
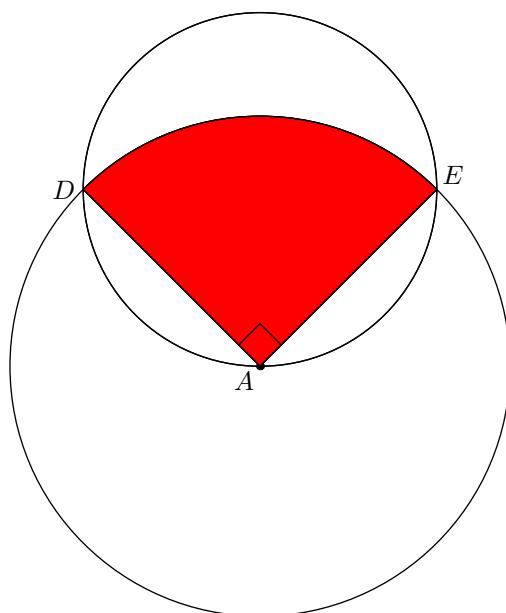
We first find the area of the intersection of ω_A and ω_B . This area is equal to the area of the intersection of ω_A and ω_C by symmetry. Let ω_A and ω_B intersect at D and E , as shown below.



Since $AD = AE = 1$ and $BD = BA = BE = \frac{\sqrt{2}}{2}$, $\angle DBA = \angle ABE = 90^\circ$ and $\angle BAD = \angle ADB = \angle BAE = \angle BEA = 45^\circ$.

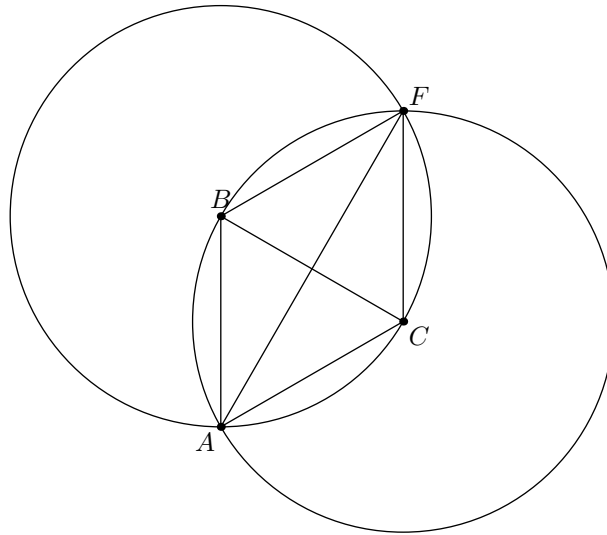
Now, $\angle DAE = 90^\circ$, so the area of the red area below is $\frac{\pi}{4}$, and the area of each green region is $\frac{1}{4} \cdot \left(\frac{\sqrt{2}}{2}\right)^2 \pi - \frac{1}{2} \cdot \left(\frac{\sqrt{2}}{2}\right)^2 = \frac{\pi}{8} - \frac{1}{4}$, so the total green area below is $\frac{\pi}{4} - \frac{1}{2}$.

Now, the area of the intersection of ω_A and ω_B is $\frac{\pi}{4} + \frac{\pi}{4} - \frac{1}{2} = \frac{\pi}{2} - \frac{1}{2}$.

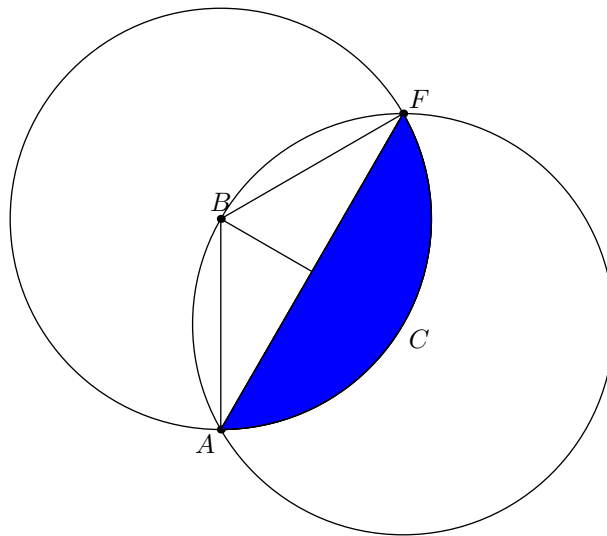


Similarly, the area of the intersection of ω_A and ω_C is $\frac{\pi}{2} - \frac{1}{2}$. We now turn our attention to the intersection of ω_B and ω_C .

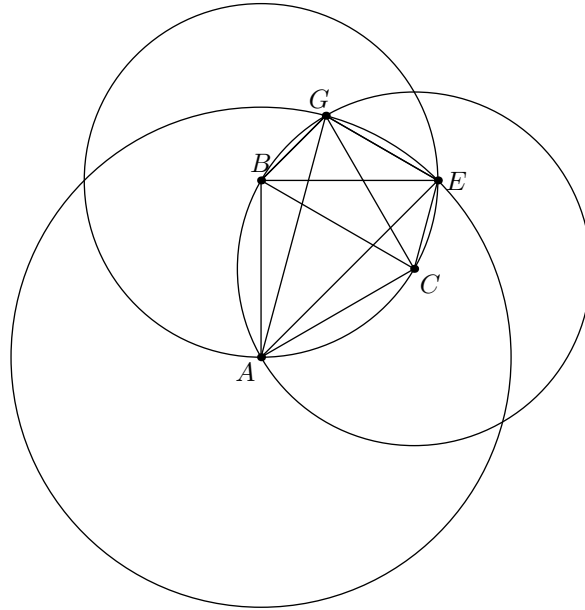
Let the intersection of ω_B and ω_C not equal to A be F.



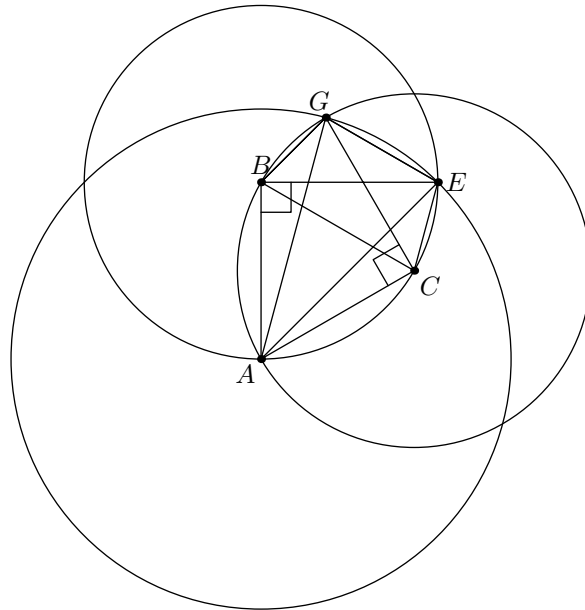
Note that $AB = BF = FC = BC = AC = \frac{\sqrt{2}}{2}$, so $\angle ABC = \angle BAC = \angle ACB = \angle CBF = \angle BFC = \angle BCF = 60^\circ$. Now, $\angle ABF = 120^\circ$, so the area of the blue region below is $\frac{1}{3} \cdot \pi \left(\frac{\sqrt{2}}{2} \right)^2 - [ABF] = \frac{\pi}{6} - \frac{\sqrt{3}}{8}$, and the intersection of ω_B and ω_C is composed of two of these blue regions, so the area of the intersection is $2 \left(\frac{\pi}{6} - \frac{\sqrt{3}}{8} \right) = \frac{\pi}{3} - \frac{\sqrt{3}}{4}$. (There are several easy ways to compute $[ABF]$ so I won't describe them here)



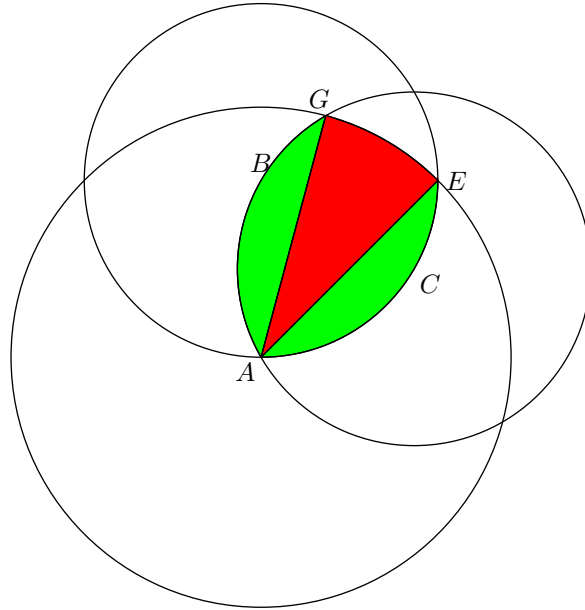
We now simply need to calculate the area of the intersection of all three circles. Again, let E be the intersection of ω_A and ω_B as earlier, and let G be the intersection of ω_A and ω_C shown below.



Observe that $AB = BE = AC = GC = \frac{\sqrt{2}}{2}$ and $AE = AG = 1$, so $\angle ABE = \angle ACG = 90^\circ$, and $\angle BAE = \angle BEA = \angle CAG = \angle CGA = 45^\circ$. Now, $\angle BAG = \angle BAC - \angle CAG = 15^\circ = \angle BAC - \angle BAE = \angle EAC$. Now, $\angle GAE = \angle BAC - \angle BAG - \angle CAE = 60^\circ - 15^\circ - 15^\circ = 30^\circ$.



Now, the area of each green area below is $\frac{1}{4} \cdot \pi \cdot \left(\frac{\sqrt{2}}{2}\right)^2 - \left(\frac{\sqrt{2}}{2}\right)^2 \cdot \frac{1}{2} = \frac{\pi}{8} - \frac{1}{4}$, and the red area is a 30° sector of ω_A , so the red area has area $\frac{\pi}{12}$. Thus, the area of the intersection of the circles is $2 \left(\frac{\pi}{8} - \frac{1}{4}\right) + \frac{\pi}{12} = \frac{\pi}{4} - \frac{1}{2} + \frac{\pi}{12} = \frac{\pi}{3} - \frac{1}{2}$.



Now, by PIE, the area of the union of the circles is $2\pi - 2\left(\frac{\pi}{2} - \frac{1}{2}\right) - \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4}\right) + \left(\frac{\pi}{3} - \frac{1}{2}\right) = 2\pi - \pi + 1 - \frac{\pi}{3} + \frac{\sqrt{3}}{4} + \frac{\pi}{3} - \frac{1}{2} = \pi + \frac{1}{2} + \frac{\sqrt{3}}{4} = \boxed{\frac{4\pi + 2 + \sqrt{3}}{4}}$.

Problem 8. Let $a \neq 1$ be the unique positive integer for which

$$\log_3(\log_3(2025 \log_a 3^a) + 3)$$

is also an integer. Compute the number of positive divisors of a .

Proposed by: Samuel Kretschmar

Answer: 76

Solution: Let $n = \log_3(\log_3(2025 \log_a 3^a) + 3)$. Then

$$\begin{aligned} 3^n - 3 &= \log_3(2025 \log_a 3^a) \\ \implies 3^{3^n-3} &= 2025 \log_a 3^a = \frac{2025a}{\log_3 a}. \end{aligned}$$

For a and $\frac{2025a}{\log_3 a}$ to be integers, $a = 3^b$ for integer b , with $\frac{2025a}{\log_3 a} = \frac{2025 \cdot 3^b}{b} = 3^{3^n-3}$. For $\frac{2025 \cdot 3^b}{b}$ to be a power of 3, we must have $b = 25 \cdot 3^d$ and

$$\frac{81 \cdot 3^{25 \cdot 3^d}}{3^d} = 3^{4+25 \cdot 3^d - d} = 3^{3^n-3}.$$

Thus $4 + 25 \cdot 3^d - d = 3^n - 3$; the only working pair is $(d, n) = (1, 4)$, not too hard to show actually with bounds. Thus, $a = 3^{25 \cdot 3^1} = 3^{75}$, which has 76 divisors.
